

1. Variables in C

A variable is a name of a memory location, container that is used to store a value.

It can store different types of data such as int, float, char, etc.

Rules for Naming Variables in C:

1. Must begin with a letter (A-Z or a-z) or an underscore _.

- Correct: total, _count

- Incorrect: 1number

2. Only letters, digits, and underscores are allowed.

- Correct: total_marks

- Incorrect: total-marks or total marks

3. No special characters like @, #, !, etc.

4. No spaces allowed.

- Correct: totalMarks

- Incorrect: total Marks

5. C is case-sensitive.

2. Data Types in C

In C language, data types are classified as:

1. Basic (Primitive)

2. Derived

3. Enumeration

4. Void

Basic data types in C include:

- int: Stores integer values

Example: `int age = 20;`

- float: Stores decimal numbers

- char: Stores single character

- double: Stores large floating-point numbers

Derived data types: Arrays, Pointers, Structures, etc.

Enumeration: Used to define named constants

Void: Represents absence of value

3. Operators in C

Operators are symbols used to perform operations on variables and values.

Types of Operators:

- Arithmetic Operators: `+`, `-`, `*`, `/`, `%`

- Relational Operators: `==`, `!=`, `>`, `<`, `>=`, `<=`

- Logical Operators: `&&`, `||`, `!`

- Assignment Operators: `=`, `+=`, `-=`, `*=`, `/=`

- Increment/Decrement Operators: `++`, `--`

- Bitwise Operators: `&`, `|`, `^`, `~`, `<<`, `>>`